

Aditya Rao

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EDUCATION

McMaster University

B.Eng in Software Engineering (CO-OP)

Hamilton, ON

Sept 2023 – Present

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math, Software Design, DSA, Computer Architecture

EXPERIENCE

Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Developing WebSphere Application Server v8.5.5, v9.0, WebSphere Liberty, and OpenLiberty and the public and internal tools which support development.
- Developed an upload feature in our translation tool which allows developers to upload translations for different languages and regions.
- Modernized the translation tool to leverage the latest technologies on OpenLiberty using Jakarta EE CDI, Restful, Concurrency, and MicroProfile Config, Telemetry, containerizing the application, and added Unit and Integration tests to ensure correctness of the workflow
- Added Apache to the translation tool as the reverse proxy (for both frontend and backend) to support TLS certificate management and OAuth authentication replacing the existing LDAP Basic Authentication
- Optimized the core data processing tool within this suite, replacing heavy framework overhead with custom API wrappers to reduce execution time by 99% (from 10 mins to <4 secs).
- Re-engineered the other 3 legacy internal tools from scratch, modernizing UX, using the latest Spring Boot and Jakarta EE technologies as above, standardizing Docker deployments, and establishing the first Unit/Integration testing pipelines to eliminate manual verification.
- Streamlined the installation FAT suite for WebSphere Application Server by removing complex infrastructure dependencies, significantly reducing debugging overhead and simplifying long-term maintenance.
- Diagnosed and solved >50 build breaks and supported the development team in diagnosing customer cases and build breaks on all platforms, including but not limited to Linux, Windows, MacOS, IBM i, and zOS
- Solved a customer case by replicating a customers environment by created E2E tests for deploying Liberty on OpenShift and verifying behavior such as Infinispan and Keycloak.
- Created a tool used by over 400+ developers and managers which allows developers to record the distribution of their work they perform and allow managers to record it and use the data to assist the developers by allocating resources in their workflows.
- Created an AI Agent leveraging IBM Bob Shell to analyze build breaks given the build break issue and description, build break logs and server configuration, OpenLiberty and WebSphere Liberty source code reducing time spent on fixing build breaks by at least 2 days worth of analysis.
- Completed a feature which allows Modernized Runtime Extensions for Java to add custom configuration actions to their WebSphere Application Server Network Deployment instances to run on server startup for Profile Conformance
- Completed a feature which allows developers to install versionless features using installUtility on WebSphere Liberty
- Cleaned up stale install issues on the OpenLiberty Github and WebSphere Liberty
- Offloaded, trained and transferred the translation tool to the build team in Kochi for easier maintainence (since they're the Client Zero for this tool)
- Led modernization of legacy L3 tools (Retain mainframe, EJB/Servlets, Ruby) to React, Spring Boot, Quarkus, and MicroProfile REST microservices, completing a multi-year Salesforce CSP migration.
- Generated knowledge transfer documents and trained the incoming intern to take over responsibilities and continue maintainence on these internal tools

Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Developed and maintained React and TypeScript code for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.
- Developed real-time doctor-patient voice-based transcription with gRPC and Firebase, achieving sub-500ms streaming latency by load balancing OpenAI Whisper across self-hosted and managed instances.
- Developed AI diagnostic hooks to analyze transcripts, highlighting clinical insights for attending physicians.

PROJECTS

- WebCraft** |  | *C++, CMake, Vcpkg, Asynchronous Programming, Github Actions*

Apr 2025 – Sept 2025

 - Engineered an asynchronous programming library in C++ 23, featuring coroutine-based task management, concurrent streams, and high-performance networking/file I/O modules to simplify cross-platform async development.
 - Automated testing and release workflows using gtest (130+ tests), GitHub Actions CI/CD, and published a custom vcpkg port for standardized cross-platform distribution.
- Resume Builder** |  | *Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker*

Aug 2024 – Feb 2025

 - Built an AI-powered resume builder with Next.js, Spring Boot, and FastAPI for real-time editing and version control.
 - Integrated LangChain to automate ATS keyword optimization, intelligent scoring, and template tailoring.
- FashioNova** |   | *Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask*

Oct 2024 – Oct 2024

 - Built a virtual wardrobe & AI fashion advisor in 36 hours at Hack the Valley 9 (UofT Scarborough).
 - Developed an OpenCV, Flask, and Socket.io pipeline for video clothing overlays with Cloudflare AI outfit suggestions.

EXTRA CURRICULARS

- Sys-Admin and Fullstack Developer Executive**

Nov 2024 – Present

IEEE McMaster Student Branch
Hamilton, ON

 - Set up an RPi 3B+ Cluster for a server to handle up to 100 clients in an instant with a budget of \$5000
 - Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster
 - Developing the website for the IEEE student branch and the admin dashboard for executives to manage

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C#, C++, SQL, C, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP
Developer Tools: AWS, GCP, Git, Docker, OpenShift, Kubernetes, Infinispan, IBM Bob
Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, Mongoose

CERTIFICATIONS

- Practical OpenShift for Developers - OpenShift 4** – *Udemy*

August 2025
- [NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025** – *Udemy*

February 2025
- Docker & Kubernetes: The Practical Guide [2025 Edition]** – *Udemy*

January 2025